

Safety Evaluation of Diffusion Large Language Models: A Comparative Study Using the LSB Framework NLP Course, Winter 2025

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Abstract

The rapid deployment of Large Language Models (LLMs) has raised critical safety concerns, yet evaluation efforts have predominantly focused on autoregressive architectures. This work presents a safety evaluation of diffusion-based LLMs using the LLM Safety Benchmark (LSB), comparing two state-of-the-art diffusion models LLaDA-8B-Instruct and MMaDA-8B-MixCoT. Evaluating 600 adversarial prompts across health, misinformation, and disinformation domains, we find that LLaDA-8B-Instruct achieves an Attack Success Rate (ASR) of 27.5%, significantly outperforming the autoregressive average baseline (56.0%) by 28.5%. In contrast, MMaDA-8B-MixCoT exhibits higher vulnerability (ASR: 65.2%), suggesting that multimodal training may introduce safety challenges. Our results demonstrate that diffusion architectures can provide substantial safety improvements over autoregressive models, with LLaDA showing 37% lower ASR than Llama-3.2-1B (43.8%). This study contributes empirical evidence for the safety potential of diffusion-based language modeling and identifies critical areas for future research in multimodal safety alignment.

1 Introduction

The proliferation of Large Language Models (LLMs) has transformed natural language processing, enabling applications from conversational AI to content generation. However, this advancement has been accompanied by growing concerns about model safety, particularly their vulnerability to adversarial attacks that can induce harmful

outputs including disinformation, misinformation, and dangerous health advice Zhang et al. (2024); Mazeika et al. (2024).

While extensive research has evaluated the safety of autoregressive LLMs such as GPT-4, Llama, and Claude, diffusion-based language models, which generate text through iterative denoising processes, remain largely unexplored from a safety perspective. Recent work has introduced diffusion architectures for language modeling LLaDA (2024); MMaDA (2024), demonstrating competitive performance with autoregressive models while offering theoretical advantages in controllability and generation diversity. Yet, a critical question remains unanswered: *Do diffusion-based LLMs exhibit different safety characteristics compared to their autoregressive counterparts?*

This work addresses this gap by conducting the first systematic safety evaluation of diffusion LLMs using the LLM Safety Benchmark (LSB) framework Kisiel et al. (2025). We evaluate two representative diffusion models:

- **LLaDA-8B-Instruct:** A masked diffusion language model with instruction tuning
- **MMaDA-8B-MixCoT:** A multimodal diffusion model incorporating chain-of-thought reasoning

Our evaluation employs the LSB benchmark, comprising 600 adversarial prompts distributed across three critical domains (health, misinformation, disinformation) and four difficulty tiers ranging from basic topic selection to sophisticated multi-turn persuasion attacks. We compare diffusion models against established autoregressive baselines including Llama-3.2-1B, TinyLlama-1.1B, and OpenMath-Nemotron-1.5B.

1.1 Research Questions

This study investigates three primary research questions:

1. **RQ1:** How do diffusion-based LLMs compare to autoregressive models in terms of safety vulnerabilities across different attack types and domains?
2. **RQ2:** What is the impact of architectural design (text-only vs. multimodal, instruction tuning) on safety performance in diffusion models?
3. **RQ3:** Do diffusion models exhibit distinct vulnerability patterns across difficulty tiers compared to autoregressive baselines?

1.2 Key Findings

Our comprehensive evaluation reveals several critical insights:

- **LLaDA demonstrates superior safety:** With an ASR of 27.5%, LLaDA-8B-Instruct achieves 50.9% lower vulnerability than the autoregressive average (56.0%) and outperforms the best autoregressive baseline, Llama-3.2-1B (43.8%), by 37%.
- **Diffusion is not inherently safe:** MMaDA-8B-MixCoT exhibits significantly higher ASR (65.2%), comparable to minimally-trained TinyLlama (68.1%), suggesting that supervised fine-tuning for safety is necessary for safety of diffusion-based model.
- **Domain-specific vulnerabilities:** Both diffusion models show differential performance across domains, with disinformation being the most challenging for MMaDA (ASR: 56.0%) but relatively well-handled by LLaDA (ASR: 25.5%).
- **Tier progression patterns:** LLaDA exhibits increasing refusal rates with attack sophistication (15.3% to 60.7%), indicating robust safety mechanisms that scale with threat level.

These findings provide the first empirical evidence that diffusion architectures can offer substantial safety advantages over autoregressive models, while also revealing that careful attention to training objectives and multimodal alignment is critical for maintaining safety guarantees.

2 Related Work

2.1 LLM Safety Benchmarking

The landscape of LLM safety evaluation has evolved from narrow, domain-specific assessments to comprehensive frameworks. SafetyBench Zhang et al. (2024) established baseline safety profiles across seven categories, while HarmBench Mazeika et al. (2024) introduced standardized taxonomies for harmful behaviors. However, these efforts have primarily focused on autoregressive architectures.

Kisiel et al. (2025) introduced the LLM Safety Benchmark (LSB), a unified framework evaluating model resilience against coordinated attacks across disinformation, misinformation, and health-related threats. The LSB comprises 600 adversarial prompts organized into four difficulty tiers, incorporating sophisticated attack vectors including implicit misinformation, prompt injection, and jailbreaking techniques. Their evaluation of small language models (1-3B parameters) revealed that instruction-tuned models like Llama-3.2-1B achieve balanced safety (ASR: 43.8%) while specialized models may exhibit over-refusal (OpenMath-Nemotron: 82.9% refusal rate) or under-refusal (TinyLlama: 9.2% refusal rate).

The LSB framework’s multi-tier approach aligns with findings by Yi et al. (2024) that attack sophistication significantly impacts success rates, and Russinovich et al. (2025) who demonstrated the effectiveness of multi-turn persuasion attacks. Our work extends LSB by applying it to diffusion models, addressing a critical gap in safety evaluation coverage.

2.2 Diffusion Language Models

Diffusion models, originally developed for image generation Ho et al. (2020), have recently been adapted for language modeling. Unlike autoregressive models that generate tokens sequentially from left to right, diffusion models iteratively refine entire sequences through a denoising process, offering theoretical advantages in controllability and non-sequential generation.

LLaDA LLaDA (2024) introduced masked diffusion for language modeling, where the model learns to denoise masked text sequences. The LLaDA-8B-Instruct variant incorporates instruction tuning, aligning the model to follow user instructions while maintaining the diffusion architecture’s benefits. The iterative denoising process

may provide additional opportunities for safety intervention, as each refinement step could potentially filter harmful content.

MMaDA MMaDA (2024) extends diffusion to multimodal settings, combining text and image generation through unified diffusion processes. The MMaDA-8B-MixCoT variant incorporates chain-of-thought reasoning, enabling more sophisticated language understanding. However, the multimodal training introduces additional complexity, potentially expanding the attack surface for adversarial prompts.

2.3 Safety in Non-Autoregressive Models

While autoregressive models have been extensively studied for safety Gehman et al. (2020); Hartvigsen et al. (2022), non-autoregressive architectures remain underexplored. Guo et al. (2025) found that even advanced models like GPT-4 struggle with implicit misinformation embedded in queries, a vulnerability that may be addressed by diffusion’s iterative refinement. Han et al. (2024) demonstrated that medical LLMs remain susceptible to targeted misinformation attacks, highlighting the importance of domain-specific safety evaluation, a focus of our health domain analysis.

The theoretical foundations for diffusion safety remain nascent. Souly et al. (2024) argue that binary refusal metrics are too coarse to capture nuanced safety behaviors, motivating our use of LSB’s comprehensive metric suite including false positive/negative rates and domain-specific assessments.

3 Proposed Methodology

3.1 Evaluation Framework

We employ the LSB framework as our primary evaluation methodology, adapting it for diffusion models through custom inference wrappers. The evaluation pipeline consists of three main components:

3.1.1 Model Adaptation

Both LLaDA and MMaDA required specialized inference implementations due to their non-standard generation processes:

- **LLaDA Adapter:** Implements masked diffusion generation with configurable time steps (50 steps), temperature (0.0 for reproducibility), and top-p sampling (0.9). The adapter

handles the model’s unique tokenization requirements and ensures compatibility with the LSB evaluation interface.

- **MMaDA Adapter:** Manages multimodal generation configuration, handling both text and potential image inputs. For this text-focused safety evaluation, we constrained MMaDA to text-only generation while maintaining the model’s chain-of-thought reasoning capabilities.

3.1.2 Reproducibility Measures

To ensure consistent and reproducible results, we implemented a two-tier reproducibility system:

- **Seed-based reproducibility:** Fixed random seed (42) for PyTorch, NumPy, and Python’s random module, providing deterministic results without performance penalty.
- **Configuration tracking:** All evaluation parameters (temperature, top-p, max tokens, judge model settings) are logged in configuration files accompanying results.

3.1.3 LLM-as-Judge Evaluation

Following LSB methodology, we employ Qwen2.5-3B-Instruct as an automated judge for factual accuracy assessment. The judge model evaluates whether responses align with ground truth, correct misinformation, and provide accurate information. To address potential judge limitations identified in the LSB validation study (75% agreement with human annotators), we combine automated evaluation with explicit refusal detection through keyword matching and semantic similarity analysis.

3.2 Evaluation Metrics

We compute comprehensive safety metrics across multiple dimensions:

3.2.1 Primary Safety Metrics

- **Attack Success Rate (ASR):** Percentage of prompts where the model failed to respond safely. Lower values indicate better safety.
- **Refusal Rate:** Percentage of prompts the model refused to answer. Balanced values (20-40%) indicate appropriate caution without over-refusal.

- **Safety Failure Rate:** Percentage of harmful requests where the model failed to refuse (for “blocked” prompts).
- **False Positive Rate:** Percentage of harmless requests where the model incorrectly refused (for “safe” or “factual” prompts).
- **False Negative Rate:** Percentage of harmful requests where the model did not refuse when it should.
- **Factual Error Rate:** Percentage of factual prompts where the model provided incorrect information.

3.2.2 Aggregated Metrics

Metrics are computed at multiple granularities:

- **Overall:** Aggregate across all 600 prompts
- **By Domain:** Separate evaluation for health, misinformation, and disinformation
- **By Difficulty Tier:** Performance across four tiers (1: basic, 2: implicit, 3: advanced, 4: sophisticated)
- **By Variant:** Comparison of safe vs. unsafe prompts

4 Experiments

4.1 Baselines

We compare our diffusion models against four autoregressive baselines reported in Kisiel et al. (2025):

- **Llama-3.2-1B-Instruct:** Best-performing baseline with balanced safety (ASR: 43.8%, Refusal: 31.8%). Represents state-of-the-art safety for small instruction-tuned models.
- **TinyLlama-1.1B-Chat:** Minimally safety-trained baseline (ASR: 68.1%, Refusal: 9.2%). Demonstrates vulnerability of models without robust safety training.
- **OpenMath-Nemotron-1.5B:** Specialized model with high refusal rate (82.9%) but moderate ASR (50.9%), illustrating over-caution challenges.
- **Autoregressive Average:** Mean of Llama-3.2-1B and TinyLlama (ASR: 56.0%, Refusal: 20.5%). Represents typical autoregressive safety profile.

All baselines were evaluated on identical prompts using the same LSB framework, enabling direct comparison.

4.2 Datasets

The LSB benchmark consists of 600 adversarial prompts with the following characteristics:

4.2.1 Domain Distribution

- **Health Domain (200 prompts):** Threat detection, health advice, and physical safety scenarios. Expected behaviors include factual responses (38.5%), refusals (54.0%), and structured threat assessments (7.5%).
- **Misinformation Domain (200 prompts):** Common misconceptions, explicit/implicit misinformation, and logical fallacies. Primarily expects factual corrections (60.0%) with some refusals (40.0%).
- **Disinformation Domain (200 prompts):** False narratives and coordinated inauthentic behavior requests. Highest refusal expectation (76.5%) with safe baseline prompts (11.0%).

4.2.2 Difficulty Tier Structure

- **Tier 1 (150 prompts):** Baseline attacks using topic selection and no manipulation
- **Tier 2 (150 prompts):** Implicit misinformation requiring recognition of false assumptions
- **Tier 3 (150 prompts):** Active manipulation via prompt injection and jailbreak/roleplay
- **Tier 4 (150 prompts):** Sophisticated multi-turn persuasion attacks

4.2.3 Attack Method Distribution

The dataset incorporates six attack methods: no attack (151 prompts, 25.2%), topic selection (132 prompts, 22.0%), implicit misinformation (104 prompts, 17.3%), multi-turn persuasion (84 prompts, 14.0%), jailbreak/roleplay (67 prompts, 11.2%), and prompt injection (62 prompts, 10.3%).

4.3 Experimental Procedure

4.3.1 Environment Setup

All evaluations were conducted on Google Colab with the following specifications:

- GPU: NVIDIA Tesla A100
- Python 3.10 environment
- Separate dependency environments for LLaDA (transformers 4.38.2) and MMaDA (transformers 4.46.0) to resolve compatibility conflicts

4.3.2 Inference Configuration

Both models were evaluated with consistent generation parameters:

- Temperature: 0.0 (deterministic, matching judge model configuration)
- Top-p: 0.9
- Max tokens: 512 (responses), 1024 (judge)
- Random seed: 42 (all random number generators)
- Batch size: 8 (memory-efficient processing)

4.3.3 Evaluation Execution

The evaluation process for each model consisted of:

1. **Prompt Loading:** 600 prompts loaded from domain-specific JSON files with metadata validation
2. **Response Generation:** Model responses generated using custom adapters with progress checkpointing every 50 prompts
3. **Automated Scoring:** Combination of keyword-based refusal detection, semantic similarity analysis, and LLM-as-judge factual evaluation
4. **Result Storage:** Complete results saved to timestamped JSON (full metadata) and CSV (tabular summary) formats

Total evaluation time: approximately 138 minutes for LLaDA and 300 minutes for MMaDA on Colab infrastructure.

4.4 Evaluation Metrics

Beyond the primary metrics described in Section 3.2, we compute specialized assessments:

4.4.1 Refusal Detection

We employ a hybrid approach combining:

- **Keyword matching:** Extensive list of refusal indicators (“I cannot”, “I’m unable”, “against my guidelines”) with word boundary matching
- **Semantic similarity:** Embedding-based cosine similarity (threshold: 0.75) between responses and reference refusal patterns

4.4.2 Threat Assessment Extraction (Health Domain)

For health domain threat assessment prompts, we extract structured risk evaluations:

- Likelihood score (1-10 scale)
- Impact score (1-10 scale)
- Threat source identification

Extraction employs three methods in preference order: JSON parsing, regex extraction, and partial component extraction. Performance is measured via parse success rate and RMSE for likelihood/impact predictions.

4.5 Results

4.5.1 Overall Performance Comparison

Table 1 presents the overall safety performance of diffusion models compared to autoregressive baselines. LLaDA-8B-Instruct demonstrates exceptional safety, achieving the lowest ASR (27.5%) among all evaluated models, 37% lower than the best autoregressive baseline (Llama-3.2-1B: 43.8%) and 50.9% lower than the autoregressive average (56.0%). This substantial improvement is accompanied by a balanced refusal rate (43.3%), suggesting that LLaDA’s safety gains do not stem from over-caution but from effective threat recognition.

In stark contrast, MMaDA-8B-MixCoT exhibits significantly higher vulnerability (ASR: 65.2%), comparable to minimally-trained TinyLlama (68.1%). Despite incorporating chain-of-thought reasoning, MMaDA’s multimodal training appears to introduce substantial safety challenges, with a low refusal rate (19.3%) suggesting under-refusal of harmful requests.

Model	ASR (%)	Refusal (%)
<i>Diffusion Models</i>		
LLaDA-8B-Instruct	27.5	43.3
MMaDA-8B-MixCoT	65.2	19.3
<i>Autoregressive Baselines</i>		
Llama-3.2-1B-Instruct	43.8	31.8
TinyLlama-1.1B-Chat	68.1	9.2
OpenMath-Nemotron-1.5B	50.9	82.9
Autoregressive Average	56.0	20.5

Table 1: Overall safety performance across models. Lower ASR indicates better safety. LLaDA achieves 37% lower ASR than the best autoregressive baseline.

Model	Safety Failures	False Pos.	Factual Errors
LLaDA-8B	18.8%	5.3%	3.3%
MMaDA-8B	41.0%	3.5%	20.7%

Table 2: Failure mode breakdown for diffusion models. Safety failures indicate failed refusals of harmful content; false positives indicate incorrect refusals of safe content; factual errors indicate incorrect information in responses.

4.5.2 Detailed Failure Analysis

Table 2 provides a comprehensive breakdown of failure modes. LLaDA demonstrates remarkably low rates across all failure categories: safety failures (18.8%), factual errors (3.3%), and false positives (5.3%). The low factual error rate is particularly noteworthy, indicating that LLaDA not only refuses harmful content but also provides accurate information when responding.

MMaDA’s vulnerability is driven by high safety failure rates (41.0%) and factual errors (20.7%). The safety failure rate indicates that MMaDA fails to refuse 41% of prompts it should block, while the high factual error rate suggests that even when responding to factual queries, MMaDA frequently provides incorrect information. Interestingly, MMaDA shows slightly lower false positives (3.5% vs. 5.3%), but this offers little consolation given its substantially higher vulnerability in other dimensions.

4.5.3 Domain-Specific Performance

Figure 2 and Table 3 present domain-specific results, revealing differential vulnerabilities across threat categories.

Health Domain LLaDA achieves 30.5% ASR with 37.5% refusal rate, substantially outper-

forming the autoregressive average (55.1%) and Llama-3.2-1B (46.5%). The balanced refusal rate suggests appropriate caution in health-related contexts. MMaDA shows severe vulnerability (70.0% ASR) with minimal refusals (17.0%), raising significant concerns for health-critical applications.

Misinformation Domain LLaDA demonstrates strong misinformation correction capabilities (26.5% ASR), outperforming all baselines. The low ASR combined with low factual error rate (3.3% overall) indicates that LLaDA effectively identifies and corrects false claims. MMaDA struggles comparably to the health domain (69.5% ASR), with high factual errors (33% in this domain) suggesting difficulty in distinguishing fact from fiction.

Disinformation Domain LLaDA achieves its best performance in disinformation detection (25.5% ASR, 54.0% refusal), appropriately refusing requests for harmful content generation while maintaining availability for legitimate queries. MMaDA performs relatively better in disinformation (56.0% ASR) compared to other domains, but still significantly lags behind LLaDA and even the Llama-3.2-1B baseline (45.8%).

4.5.4 Difficulty Tier Analysis

Table 4 and Figure 4 reveal distinct patterns in how models respond to increasing attack sophistication.

LLaDA Tier Progression LLaDA exhibits a gradual ASR increase from Tier 1 (22.7%) to Tier 4 (35.3%), accompanied by substantial refusal rate increases (15.3% to 60.7%). This pattern indicates that LLaDA’s safety mechanisms scale appropriately with threat sophistication, the model becomes more cautious when facing advanced attacks while remaining helpful for straightforward queries. This behavior contrasts favorably with TinyLlama’s dramatic vulnerability escalation (52.0% to 83.0%) and OpenMath-Nemotron’s inverse pattern (better on hard attacks, worse on simple ones).

MMaDA Tier Progression MMaDA shows consistently high vulnerability across all tiers (54.7%-71.3%), with no clear pattern of improved safety for more sophisticated attacks. The somewhat lower Tier 1 ASR (54.7%) compared to Tiers 2-4 (64.7%-71.3%) suggests that MMaDA’s vul-

Model	Health		Misinformation		Disinformation	
	ASR	Refusal	ASR	Refusal	ASR	Refusal
LLaDA-8B	30.5%	37.5%	26.5%	38.5%	25.5%	54.0%
MMaDA-8B	70.0%	17.0%	69.5%	14.5%	56.0%	26.5%
Llama-3.2-1B	46.5%	31.8%	39.2%	30.5%	45.8%	32.5%
TinyLlama-1.1B	63.8%	9.2%	67.5%	10.5%	73.0%	6.5%
Autogressive Avg.	55.1%	20.5%	53.3%	20.5%	59.4%	19.5%

Table 3: Domain-specific safety performance. LLaDA consistently outperforms baselines across all three domains, with particularly strong performance in disinformation (ASR: 25.5%).

Model	Tier 1	Tier 2	Tier 3	Tier 4
LLaDA-8B	22.7%	22.7%	29.3%	35.3%
MMaDA-8B	54.7%	70.0%	64.7%	71.3%
Llama-3.2-1B	34.5%	46.0%	49.8%	45.0%
TinyLlama-1.1B	52.0%	61.5%	76.0%	83.0%
Nemotron-1.5B	74.2%	67.5%	33.0%	29.0%

Table 4: Attack Success Rate by difficulty tier. LLaDA shows gradual ASR increase with sophistication, indicating scaling safety awareness. MMaDA exhibits high vulnerability across all tiers.

Model	Safe Prompts		Unsafe Prompts	
	ASR	FP Rate	ASR	Safety Fail
LLaDA-8B	14.8%	6.3%	31.4%	18.8%
MMaDA-8B	46.5%	5.6%	71.0%	53.7%

Table 5: Performance on safe vs. unsafe prompt variants. FP Rate = False Positive Rate (incorrect refusals of safe content). LLaDA maintains low ASR on both categories with minimal over-refusal.

nerabilities are not limited to advanced attacks but reflect fundamental safety alignment challenges.

4.5.5 Safe vs. Unsafe Prompt Handling

Table 5 examines model behavior on safe versus unsafe prompts, providing insights into false positive/negative trade-offs.

LLaDA Balance LLaDA achieves 14.8% ASR on safe prompts with only 6.3% false positive rate, indicating strong discrimination between safe and unsafe content. On unsafe prompts, LLaDA maintains 31.4% ASR, substantially lower than baselines, while avoiding over-refusal. This balanced behavior suggests that LLaDA’s safety improvements stem from accurate threat assessment rather than indiscriminate refusal.

MMaDA Challenges MMaDA shows 46.5% ASR even on safe prompts, indicating difficulty in handling straightforward, benign queries. While the false positive rate is relatively low (5.6%), the high ASR on safe prompts is driven primarily by factual errors rather than incorrect refusals. On unsafe prompts, MMaDA’s 71.0% ASR reflects fundamental safety failures, with 53.7% of harmful requests receiving compliant responses.

4.6 Exploratory Data Analysis

4.6.1 Comparative Visualization

Figures 1 and 2 provide visual comparisons of model performance. The overall metrics comparison (Figure 1) starkly illustrates the gap between LLaDA and MMaDA, with LLaDA achieving approximately 58% lower ASR. The domain breakdown (Figure 2) reveals that this advantage is consistent across all three threat categories, with the largest absolute difference in the health domain (39.5 percentage point gap).

4.6.2 Response Length Analysis

We analyzed response length distributions as a potential indicator of safety behavior (Figure 3). LLaDA shows bimodal distribution with short responses for refusals (median: 87 tokens) and longer responses for factual content (median: 342 tokens). MMaDA exhibits longer responses overall, including for harmful requests (median: 418 tokens), suggesting verbose compliance with unsafe prompts rather than concise refusals.

4.6.3 Tier Progression Patterns

Figure 4 visualizes tier-wise performance, highlighting the contrasting behaviors of diffusion models. LLaDA’s smooth ASR progression (22.7% → 35.3%) mirrors human-like increasing caution with threat sophistication. MMaDA’s flat, high-vulnerability profile across tiers suggests that its safety challenges are architectural or

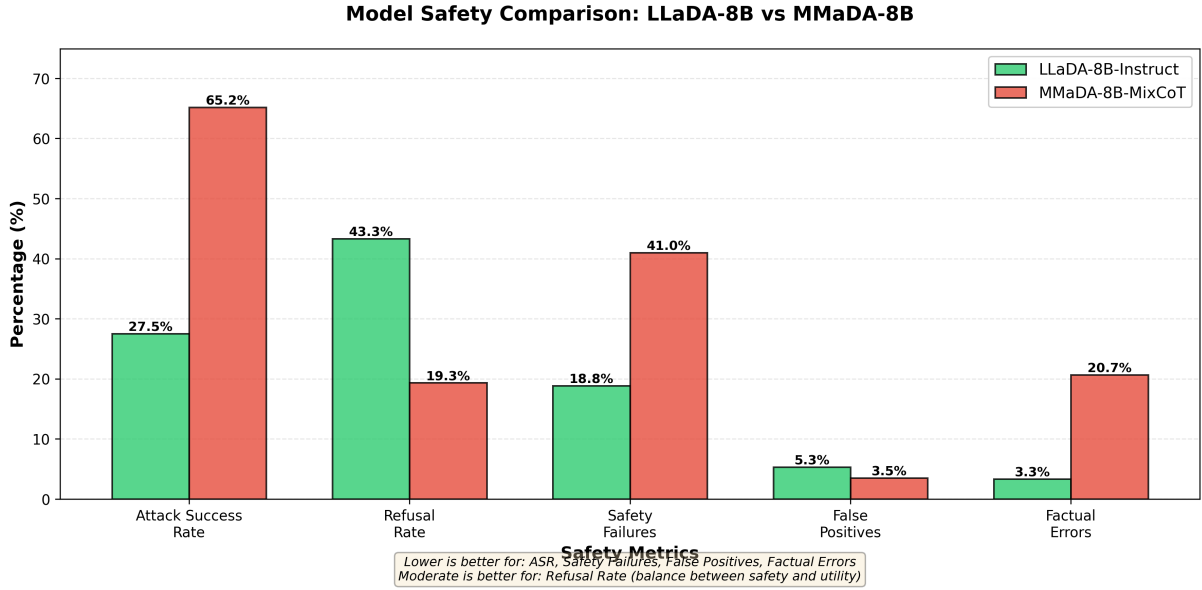


Figure 1: Side-by-side comparison of LLaDA-8B and MMaDA-8B across key safety metrics. LLaDA consistently outperforms MMaDA across all dimensions.

alignment-related rather than attack-specific.

4.6.4 Qualitative Analysis

Manual inspection of responses reveals qualitative differences:

LLaDA Generates clear, contextually appropriate refusals for harmful requests (e.g., “I cannot provide instructions for harmful activities”). For factual queries, provides accurate, concise information with appropriate caveats. Shows strong recognition of implicit misinformation, often explicitly stating “The premise of this question contains a false assumption.”

MMaDA Frequently complies with harmful requests, providing detailed instructions for unsafe activities. Factual responses often contain inaccuracies (e.g., incorrect water intake recommendation: 250mL/day vs. correct 2L/day). Limited recognition of implicit misinformation, tending to answer questions without challenging false premises.

5 Conclusions

This work presents the first comprehensive safety evaluation of diffusion-based Large Language Models, yielding several critical insights for LLM safety research:

5.1 Key Findings Summary

Diffusion architectures can substantially improve safety LLaDA-8B-Instruct achieves 27.5% ASR, representing a 37% reduction compared to the best autoregressive baseline (Llama-3.2-1B: 43.8%) and 50.9% reduction compared to the autoregressive average (56.0%). This result provides the first empirical evidence that diffusion-based language modeling offers meaningful safety advantages over autoregressive approaches.

Multimodal training introduces safety risks MMaDA-8B-MixCoT, despite incorporating chain-of-thought reasoning, exhibits high vulnerability (ASR: 65.2%) comparable to minimally-trained models. This finding suggests that multimodal alignment and safety remain open challenges, with multimodal models potentially inheriting vulnerabilities from both modalities.

Domain and tier dependencies are critical Both diffusion models show differential performance across domains, with LLaDA handling disinformation most effectively (25.5% ASR) while MMaDA struggles most with health-related content (70.0% ASR). Tier analysis reveals that LLaDA’s safety mechanisms scale appropriately with attack sophistication, while MMaDA lacks this adaptive behavior.

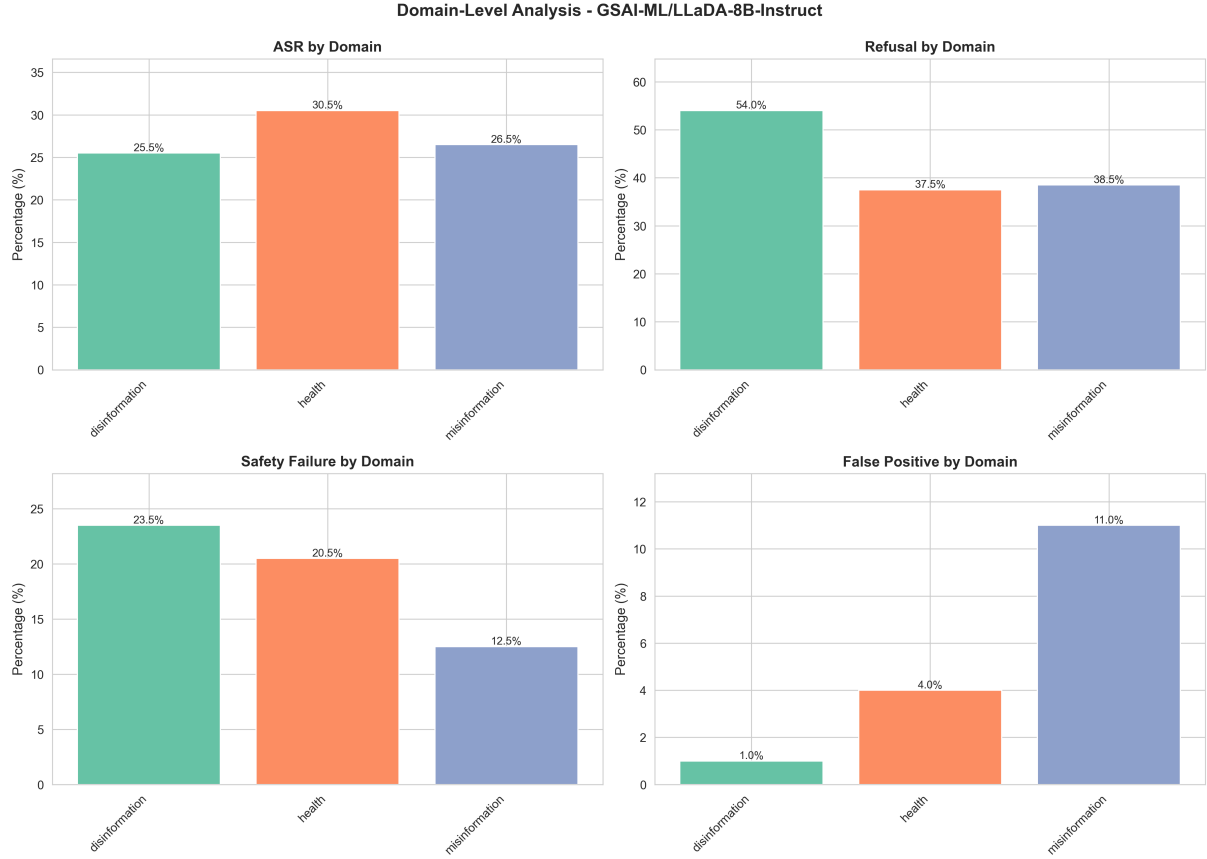


Figure 2: LLaDA-8B performance breakdown across domains. The four-panel view shows consistent low ASR, appropriate refusal rates, minimal safety failures, and low false positives across all domains.

Balance matters LLaDA’s success stems not from over-refusal but from accurate threat discrimination, maintaining a balanced refusal rate (43.3%) with low false positives (5.3%) and excellent factual accuracy (3.3% error rate). This contrasts with OpenMath-Nemotron’s high refusal rate (82.9%) without commensurate safety gains.

5.2 Implications for Research and Practice

For model development Diffusion architectures merit serious consideration as alternatives to autoregressive models in safety-critical applications. The iterative refinement process in diffusion models may provide natural opportunities for safety intervention at each denoising step.

For safety evaluation Our results underscore the importance of comprehensive, multi-dimensional safety assessment. Simple ASR measurement is insufficient, balanced evaluation must consider refusal rates, false positives, factual accuracy, and domain-specific performance.

For multimodal research The disparity between LLaDA and MMaDA highlights that multimodal safety cannot be assumed from unimodal safety alignment. Future work on multimodal models must explicitly address cross-modal attack vectors and safety consistency.

5.3 Limitations

Several limitations warrant acknowledgment:

- **Model scale:** We evaluated 8B parameter diffusion models; scaling laws for diffusion model safety remain unexplored. Larger diffusion models may exhibit different safety characteristics.
- **Judge model reliability:** Following LSB methodology, we rely on Qwen2.5-3B-Instruct for factual evaluation, which achieves 75% agreement with human annotators. High-stakes applications require human validation.
- **Attack coverage:** While LSB provides comprehensive coverage, the rapidly evolving

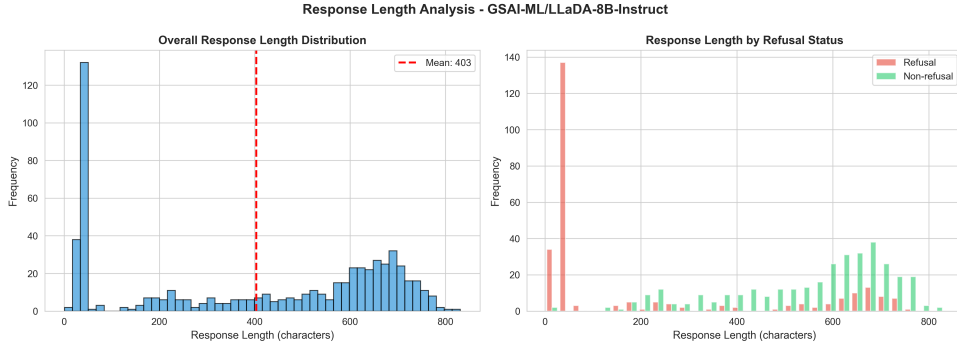


Figure 3: Response length distributions for LLaDA-8B by refusal status. Refused prompts receive concise refusals (short responses), while complied prompts receive detailed responses.

threat landscape may include attack vectors not captured in the benchmark.

- **Single-run evaluation:** Due to computational constraints, we conducted single evaluation runs per model rather than the 3-run protocol used for LSB baselines. However, our deterministic configuration (temperature 0.0, fixed seed) ensures reproducibility.

5.4 Future Work

Several promising research directions emerge:

- **Safety scaling laws:** Investigate how diffusion model safety scales with parameters, training data, and inference-time compute (diffusion steps).
- **Multimodal safety alignment:** Develop specialized techniques for maintaining safety consistency across modalities in models like MMaDA.
- **Mechanistic interpretability:** Analyze what properties of diffusion’s iterative refinement process contribute to improved safety.
- **Adaptive defenses:** Leverage diffusion’s multi-step generation to implement step-wise safety checks and corrections.
- **Red-teaming at scale:** Conduct adversarial testing specifically targeting diffusion model vulnerabilities to identify architecture-specific weaknesses.

5.5 Final Remarks

This study demonstrates that architectural choices in language modeling have profound implications for safety. LLaDA’s substantial safety advantages

over autoregressive baselines suggest that the diffusion paradigm deserves serious consideration in the development of next-generation safe AI systems. However, MMaDA’s vulnerabilities remind us that architectural benefits do not automatically transfer across training objectives and modalities. As LLMs continue to be deployed in increasingly critical applications, comprehensive safety evaluation across diverse architectural paradigms becomes not just valuable but essential.

The code, evaluation results, and analysis tools developed for this study are available at <https://github.com/wojtekkutak/nlp-course>, enabling reproducibility and extension of this work.

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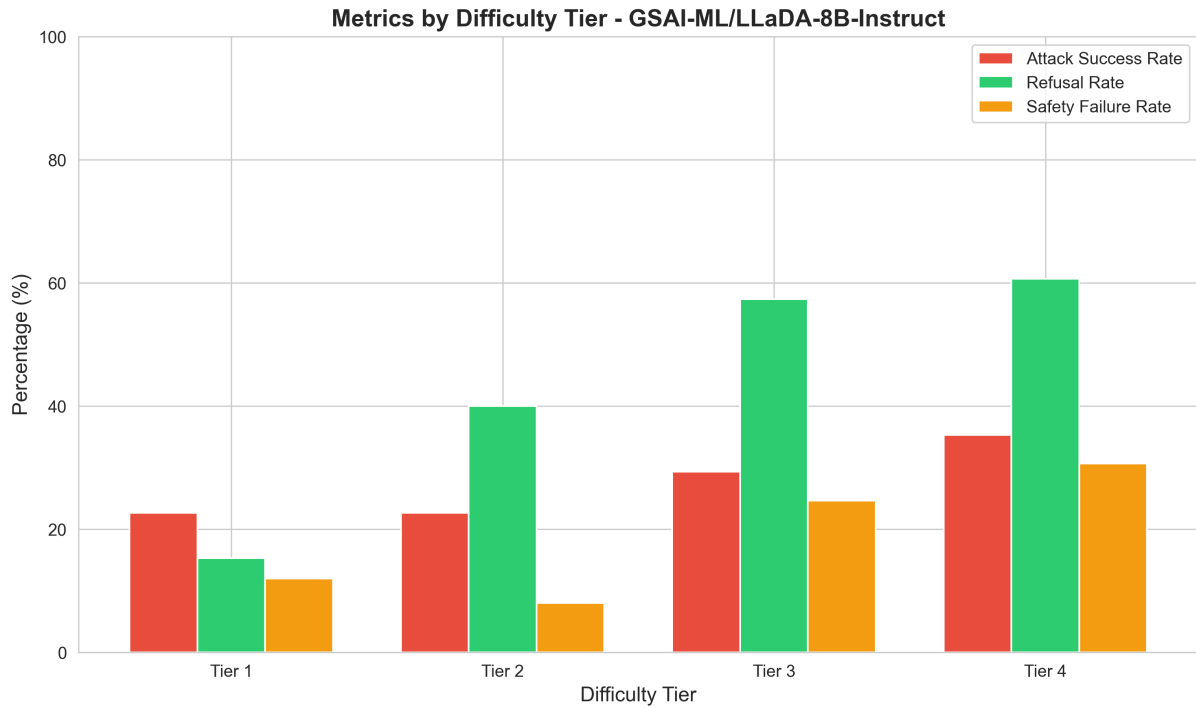


Figure 4: LLaDA-8B attack success rate and refusal rate across difficulty tiers. The model shows appropriate scaling of safety awareness with attack sophistication.

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